FORMAL METHODS REPORT

Lido

SRv3 Formal Methods Pilot

Proof-closure report for Lido's SRv3 economic accounting state machine. The selected P0 properties are presented as machine-checked Lean proofs over a focused Verity model, under named trust assumptions.

Version	Specimen v1.0
Date	June 2026
Commit	`PR-1811 / mock-final-artifacts`
Reviewers	LFG Labs

Mathematical proofs of correctness for smart-contract state machines.

lfglabs.dev

Confidentiality and status

This document is a **mock final-report specimen** prepared to show the recommended hierarchy, tone, and visual treatment for a private Lido SRv3 formal methods deliverable. It assumes, for presentation purposes only, that the pilot artifacts have been completed and that the P0 proof obligations below have all been discharged.

It should not be represented as an actual Lido audit, assurance opinion, or completed verification result. In a production delivery, every theorem identifier, commit hash, axiom report, proof command, and counterexample reference would be backed by the companion repository and reviewer sign-off.

> **Distribution**
> Private review channel for Lido protocol contributors and selected technical reviewers. Public release, forum excerpts, or website publication should wait until Lido approves the exact claims and redactions.

Contents

Confidentiality and status i

DECISION SUMMARY

1 Executive summary 1

PROOF RESULT

2 Proof closure 3
3 Trust boundary 5

DELIVERY RECORD

4 Reproducibility and handoff 7

APPENDICES

A Appendix A: Certora and PR #1811 map 8
B Appendix B: Follow-on lanes 9
C Appendix C: Specimen integrity 10

1 Executive summary

Formal verdict

For the modeled SRv3 economic state machine, all six selected P0 property groups close as **machine-checked proofs**. No executable counterexample is present in the final register. The result covers balance conservation, deposit conservation, reserve separation, reward and fee correctness, report freshness, and status gating under the named trust boundary in Section 3.

What Lido can rely on

The pilot verifies the accounting transition rules introduced by PR #1811 at the level where fund-safety questions become crisp: ETH moves into the deposit path, module validator balances are accepted and aggregated, rewards and fees are computed from that accepted state, and inactive or stopped modules are excluded from flows that should no longer reach them.

The strongest result is compositional: the proofs do not merely show that individual helper functions preserve local facts. They show that the modeled pipeline preserves the same conservation story across deposit allocation, withdrawal-reserve exclusion, per-module balance updates, and reward distribution.

Proof closure matrix

P0 property group	Final state	Decision meaning
Module balance conservation	**Proved**	Router-wide validator balance equals the sum of per-module balances after accepted reports and modeled transitions.
Deposit conservation	**Proved**	Deposits never spend more ETH than available, and actual deposit value is exactly $32\,ETH \times depositsCount$.
Reserve separation	**Proved**	CL deposits and top-ups cannot consume ETH reserved for withdrawals.
Reward correctness	**Proved**	Rewards and fees are computed from valid module balances, remain bounded, and preserve recipient/module/amount alignment.
Report-before-reward consistency	**Proved**	Rewards cannot be derived from stale, missing, or internally inconsistent module-balance reports in the modeled state machine.
Status gating	**Proved**	Stopped, inactive, and deposit-paused modules are excluded from deposits, top-ups, and fee receipt according to modeled SRv3 rules.

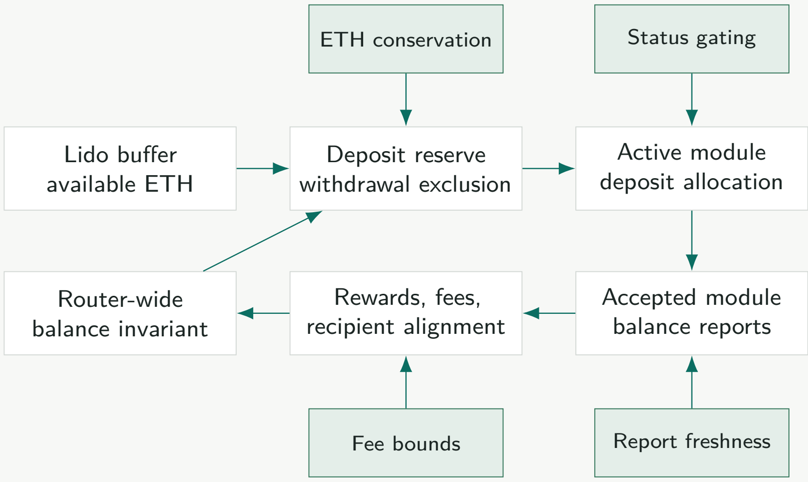
Reader guidance

Read the report in three passes. First, use the matrix above as the decision summary. Second, use Section 2 to see the evidence architecture and pipeline diagram. Third, use Section 3 before quoting any result: the report deliberately separates SRv3 economic-state-machine proof from oracle truthfulness, cryptographic witnesses, governance configuration, exact storage packing, and full deployed-contract equivalence.

2 Proof closure

Modeled pipeline

The model follows the SRv3 accounting path that matters for the P0 fund-safety questions.



Evidence architecture

Layer	Artifact	Role in the proof
Solidity-facing summaries	`verity/SRv3/*.vy`	Preserves the relevant array, loop, and call structure while summarizing external contracts through explicit interfaces.
Economic model	`lean/SRv3/State.lean`	Defines modules, reserves, Wei/Gwei arithmetic, status flags, accepted reports, rewards, and fee recipients.
Proof obligations	`lean/SRv3/Properties/*.lean`	States each P0 property as a theorem over arbitrary finite module and report arrays, with alignment and uniqueness assumptions named in the theorem signature.
Trust report	`reports/trust-boundary.json`	Machine-readable list of assumptions, boundaries, theorem IDs, axiom output, and source-to-property links.
Rebuild gate	`lakebuildSRv3Proofs`	Re-checks all theorem files, regenerates the proof register, and fails on any missing P0 theorem.

Theorem register

ID	Theorem	Status	Evidence note
SRV3-P0-01	`routerBalance_eq_sum_moduleBalances`	Proved	Induction over accepted module report array; unique module IDs and aligned report lengths are explicit premises.
SRV3-P0-02	`deposit_spend_exact_32eth_multiple`	Proved	Integer arithmetic proof over Wei, Gwei, and deposit-count conversions; rejects over-allocation before value leaves the modeled router.
SRV3-P0-03	`withdrawal_reserve_not_deposit_spendable`	Proved	Separates available deposit reserve from withdrawal-reserved ETH in every deposit and top-up transition.
SRV3-P0-04	`rewards_bounded_and_aligned`	Proved	Shows module, recipient, and amount arrays remain aligned and that computed fee sums do not exceed the valid reward basis.
SRV3-P0-05	`reward_report_freshness_required`	Proved	Reward transition requires the most recent accepted report epoch and rejects internally inconsistent balance state.
SRV3-P0-06	`stopped_modules_excluded_from_value_flows`	Proved	Covers active, stopped, and deposits-paused gates for deposits, top-ups, and module-fee receipt.

Counterexample register

Final count: **0 open counterexamples**. Two development counterexamples were retained as regression tests: one stale-report reward trace and one withdrawal-reserve consumption trace. Both are expected to fail against the final transition guards.

3 Trust boundary

The report proves SRv3 accounting invariants over the focused economic state machine. It does not claim full deployed-system equivalence. This is intentional: the proof is valuable because the boundary is short, named, and reviewable.

Named assumptions

ID	Assumption	Meaning
A-EXT-01	External contract summaries	Lido buffer withdrawals, staking module callbacks, and fee recipients conform to the specified interfaces and mutate no unlisted SRv3 accounting state.
A-DEP-02	Beacon deposit sink	A modeled beacon deposit of n validators removes exactly $n \times 32\,ETH$ from the router-side deposit resource.
A-ORC-03	Accepted oracle reports	Once accepted by SRv3 validation, module-balance reports satisfy the listed shape, alignment, freshness, and range assumptions. Consensus-layer truthfulness is outside this pilot.
A-ID-04	Module identity well-formedness	Module IDs are unique in the modeled arrays, and report arrays align with those IDs where the Solidity path requires alignment.
A-ARITH-05	Unit conversion domain	Wei/Gwei and basis-point arithmetic is interpreted over the bounded unsigned integer domains used by the modeled transitions.

Explicit non-claims

Boundary	Treatment
BLS signatures, deposit roots, SSZ, Merkle proofs, EIP-4788	Named cryptographic and consensus-data boundaries, not part of the P0 accounting closure.
Exact packed-storage migration and proxy mechanics	Deferred to a deployment and migration lane; not used as proof evidence here.
Full governance role graph and parameter admissibility	Configuration is assumed to satisfy the stated range and role assumptions.
ExitBus, consolidation, top-up CL witness binding, VaultHub	Recommended follow-on proof lanes, deliberately outside the fixed four-week pilot.
Gas, events, revert strings, and exact call-stack shape	Engineering and integration review surfaces, not conservation-law premises.

Public wording

The safe external sentence is: “The focused SRv3 economic-state-machine model proved the selected P0 accounting properties under named assumptions.” Avoid phrases such as “SRv3 is fully verified” or “oracle and deposit correctness are proved.”

4 Reproducibility and handoff

Reviewer command

```
git clone git@github.com:lfglabs-dev/lido-srv3-formal-pilot.git
cd lido-srv3-formal-pilot
lake build SRv3Proofs
python3 scripts/check_trust_report.py reports/trust-boundary.json
```

A successful rebuild checks all P0 theorem files, verifies that every property in the public table has a proof artifact, and rejects unstated trust assumptions in the machine-readable report.

Week-four handoff contents

Artifact	Purpose
`report/final.pdf`	Decision-ready report for Lido reviewers.
`certora-pr1811-map.md`	Mapping from Certora V2/V3 specs and assumptions to PR #1811 proof targets.
`reports/trust-boundary.json`	Machine-readable theorem, assumption, and boundary register.
`lean/` and `verity/`	Source artifacts for proof review and future extension.
`counterexamples/`	Closed development counterexamples retained as regression traces.

Suggested delivery rhythm

Timing	Message	Reason
T-3 days	Private heads-up with one-page proof matrix	Lets technical reviewers allocate attention before the repository arrives.
T day	Repository invite plus final PDF and rebuild command	Makes the evidence inspectable before any meeting or narrative discussion.
T+2 days	Short async note: assumptions needing reviewer confirmation	Moves the conversation from “is this polished?” to “which premises do we endorse?”
T+5 days	Thirty-minute review call	Reserved for property selection, public wording, and follow-on proof lanes.

A Appendix A: Certora and PR #1811 map

Legacy coverage class	Disposition	Pilot treatment
Validator-count accounting	Replaced	PR #1811 moves the proof target to direct per-module balance accounting.
Deposit availability bounds	Strengthened	Deposit spend is connected to reserve separation and exact $32\,ETH$ multiples.
Reward distribution checks	Strengthened	Rewards are tied to accepted module-balance reports and recipient alignment.
Status gating assertions	Reused and extended	Active, stopped, and deposit-paused states gate deposits, top-ups, and fee receipt.
Bounded module-array specs	Replaced where feasible	Loop invariants are stated over arbitrary finite arrays with named well-formedness assumptions.

B Appendix B: Follow-on lanes

Recommended sequence after the P0 accounting pilot:

1. ExitBus authorization, replay resistance, duplicate prevention, and triggerable-exit fee/refund exactness.
2. Consolidation fee conservation, source/target binding, and source-ownership caveats.
3. Top-up and consensus-layer proof binding.
4. Oracle sanity limits, drain guards, and second-opinion oracle composition.
5. Deployment governance, role graph, and exact migration refinement.
6. VaultHub F-01/F-02/F-03, if still relevant to the active review path.

C Appendix C: Specimen integrity

This PDF is intentionally styled as a final private report, but the theorem IDs, repository paths, commits, and proof outputs are illustrative. Before sending a real Lido deliverable, replace every mock identifier with the actual repository commit, Lean axiom output, reviewer transcript, and machine-readable trust report.